

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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CHEMISTRY

9729/03

Paper 3 Free Response

15 September 2017

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

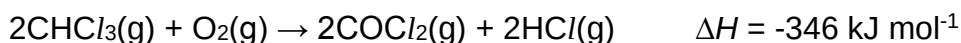
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer **all** questions in this section.

- 1 Compounds containing halogens form a large class of chemicals, which are of industrial importance and are synthesised in huge quantities worldwide.

- (a) Phosgene, COCl_2 , is a gas which was produced during World War I, and was used as a chemical weapon. One method to produce phosgene from trichloromethane is as shown below.



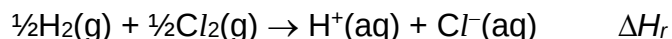
Using the data below, calculate the standard enthalpy change of formation of $\text{HCl}(\text{g})$.

standard enthalpy change of formation of $\text{CHCl}_3(\text{g})$	-103 kJ mol^{-1}
standard enthalpy change of formation of $\text{COCl}_2(\text{g})$	-184 kJ mol^{-1}

[1]

- (b) Aqueous hydrochloric acid is another chemical with many industrial uses. It can be obtained by reacting hydrogen gas and chlorine gas in the presence of UV light, followed by dissolving the hydrogen chloride gas in water.

The equation for this reaction is shown below.

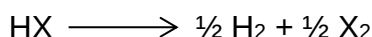


Using the Data Booklet, the data in the table below, and your answer in (a), construct an appropriate energy-level diagram to calculate ΔH_r .

	$\Delta H^\ominus / \text{kJ mol}^{-1}$
First ionisation energy of $\text{H}(\text{g})$	+1310
First electron affinity of $\text{Cl}(\text{g})$	– 364
Standard enthalpy change of hydration of $\text{Cl}^-(\text{g})$	– 381
Standard enthalpy change of hydration of $\text{H}^+(\text{g})$	– 1130

[4]

- (c) When heated, Group 17 hydrides, HX , undergo thermal decomposition.



Using the data on bond energies of the various H-X bonds given below, predict and explain how the thermal stability of the hydrides varies down Group 17.

Bonds	Bond energies/ kJ mol^{-1}
H-Cl	+431
H-Br	+366
H-I	+299

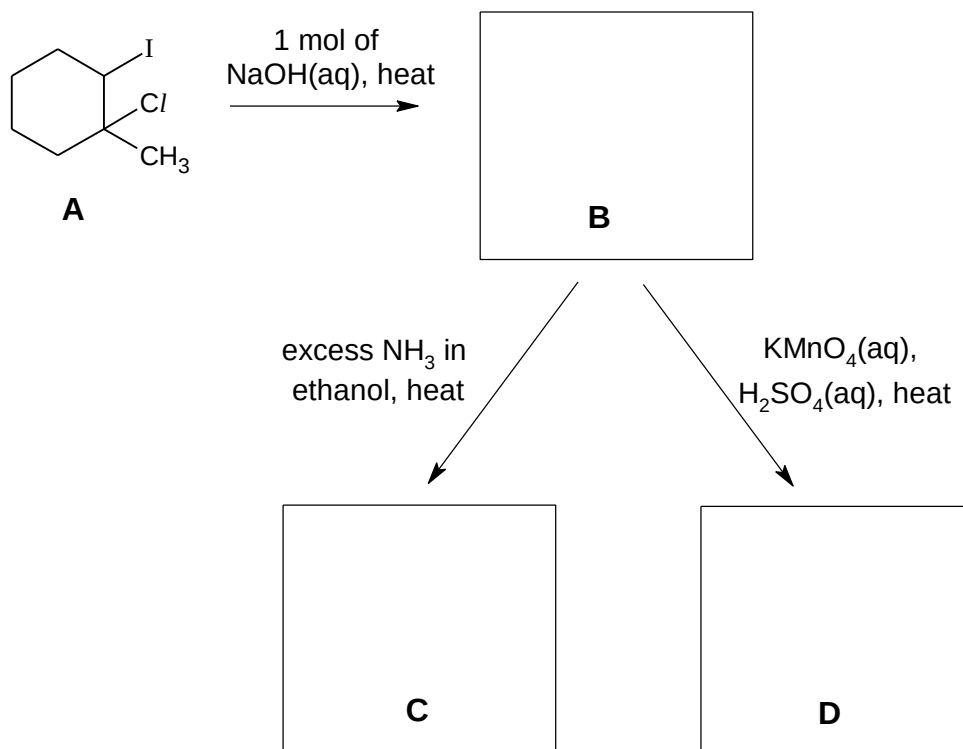
[2]

- (d) Seawater is an important source of ionic halides like sodium chloride, sodium bromide and sodium iodide.

When silver nitrate is added to separate test-tubes containing aqueous sodium chloride, sodium bromide and sodium iodide, the various AgX (X = Cl, Br and I) precipitates were observed. The solubilities of these precipitates upon addition of dilute NH₃ and concentrated aqueous NH₃, are recorded below.

	NaCl	NaBr	NaI
To NaX(aq), add AgNO ₃ (aq)	white ppt observed	cream ppt observed	yellow ppt observed
To the resulting mixture, add dilute NH ₃ (aq)	soluble	insoluble	Insoluble
To the resulting mixture, add excess concentrated aqueous NH ₃ (aq)	soluble	soluble	Insoluble

- (i) Write the balanced ionic equation for the formation of the cream precipitate from adding silver nitrate to sodium bromide. [1]
- (ii) Explain the effect of adding aqueous ammonia to each of the resulting mixture and clearly account for any difference in the observation. [3]
- (e) Halogenoalkanes are widely synthesised by reacting halogens with alkenes. In turn, halogenoalkanes can undergo nucleophilic substitution using aqueous sodium hydroxide to form alcohols commercially.
- (i) Give one reason why iodoalkanes are more reactive towards nucleophilic substitution with aqueous NaOH than the corresponding chloroalkanes. [1]
- (ii) A sequence of reactions, starting from compound **A**, a dihalogeno compound, is shown on the next page.

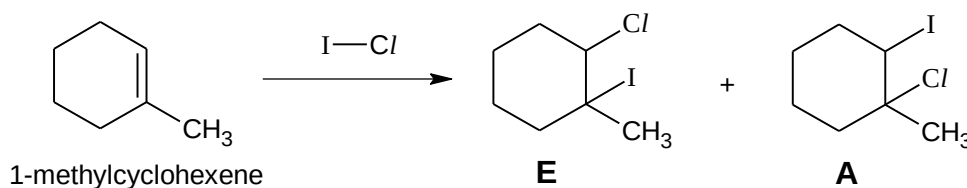


Draw the structures of compounds **B**, **C** and **D**.

[3]

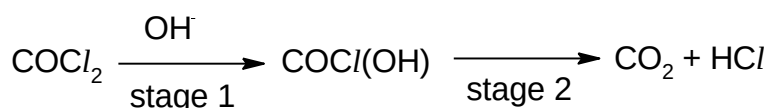
- (iii) The halogens form many interhalogen compounds, one such example is ICl , commonly known as Wijs' reagent.

Wijs' reagent can react with 1-methylcyclohexene to form a mixture of compound **E** and **A**. Explain why **A** is the major product for this reaction.



[2]

- (f) (i) Below outlines a reaction scheme of phosgene with NaOH(aq) .



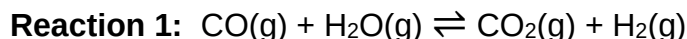
State the types of reaction occurring in stage 1, and stage 2.

[2]

- (ii) To remove any phosgene present in chloroform, a small amount of methanol is added. Use the information in (f)(i), suggest the structure of the organic compound formed when phosgene reacts with methanol. [1]

[Total:20]

- 2 (a) Carbon monoxide reacts with steam over a copper / zinc catalyst, to form a mixture of carbon dioxide and hydrogen gas.

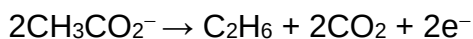


When an equimolar mixture of hydrogen and carbon dioxide at an initial total pressure of 4 atm is allowed to reach equilibrium at 1000 K, the percentage of steam in the mixture of gases is found to be 15%.

- (i) Calculate K_p of the system in **reaction 1** at 1000 K. [2]
- (ii) Suggest, with a reason, whether a higher pressure will favour the formation of steam. [1]

- (b) Sodium ethanoate can be used to prepare ethane by an electrochemical reaction, which is known as the Kolbe electrolysis reaction.

Ethane is formed at the anode by the following reaction:



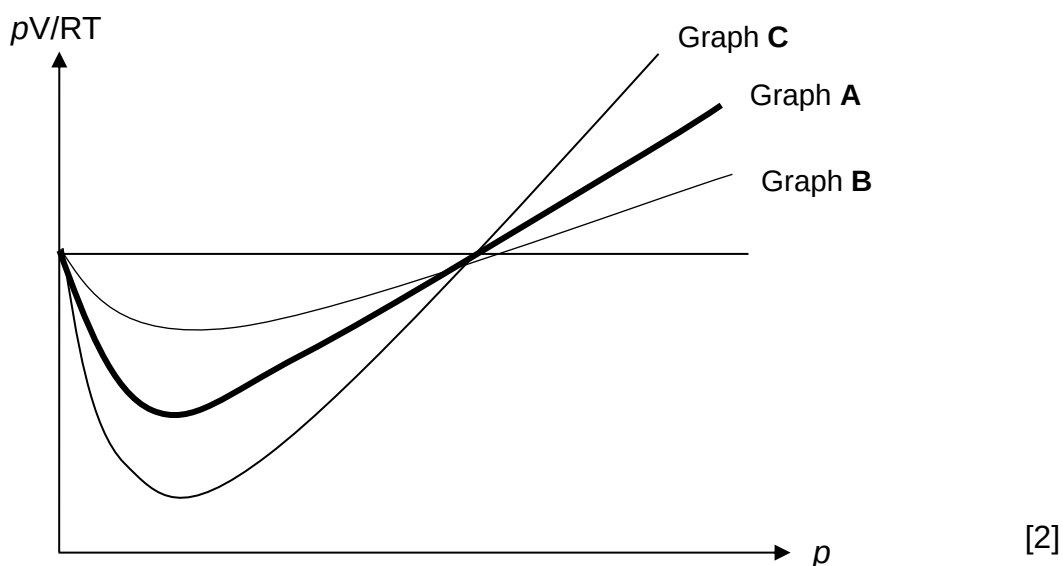
A student investigated the electrolysis of an aqueous solution of sodium ethanoate, using graphite electrodes in a litmus solution.

- (i) Give the equation for the reaction occurring at the cathode, and state the final colour expected at the cathode.

Construct an equation for the **overall** reaction. [2]

- (ii) Calculate the time, in minutes, needed by the student to pass a current of 5.0 A through a solution of sodium ethanoate, to produce 1.50 g of ethane. [2]

- (c) (i) The value of pV/RT is plotted against p for the following three gases, where p is the pressure and V is the volume of the gas. Given that graph **A** represents 1 mol of CO_2 at 298 K, identify which of the following graphs, **B** or **C**, represent 1 mol of CO_2 at 500 K. Explain your answer.



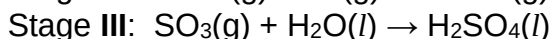
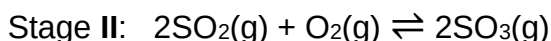
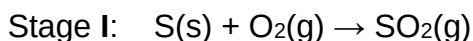
- (ii) Sulfur dioxide is used by the Romans in winemaking, when they discovered that burning sulfur candles inside empty wine vessels keeps them fresh and free from vinegar smell.

One such cylinder of sulfur dioxide has an internal volume of 2.5 dm^3 , and a mass of 2.3 kg.

Calculate the pressure (in pascals) that the sulfur dioxide would exert inside the cylinder at room temperature. [1]

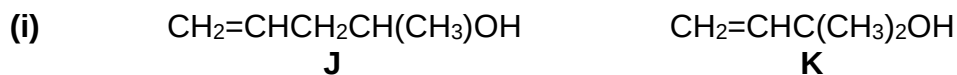
- (d) Sulfuric acid is used in many industrial processes of major importance.

The process used to produce sulfuric acid involves a three-stage process. The first stage is to pass air over burning sulfur. The emerging gas is then passed over a catalyst, V_2O_5 , which is maintained at $450\text{--}550^\circ\text{C}$ in the reaction chamber.

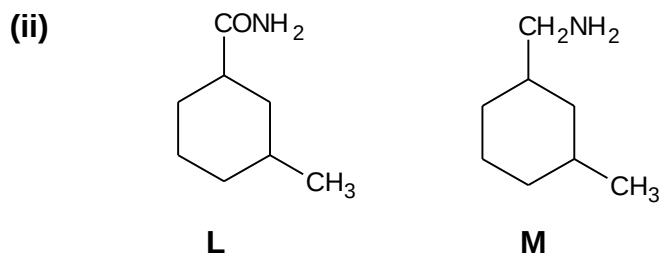


- (i) Draw dot-and-cross diagrams to show the bonding in SO_2 and H_2SO_4 . State the bond angle, with respect to sulfur atom, in each of them. [4]
- (ii) The boiling points of sulfur trioxide and sulfur dioxide are 45°C and -10°C respectively. Account for the difference in boiling points for both compounds, in terms of structure and bonding. [2]

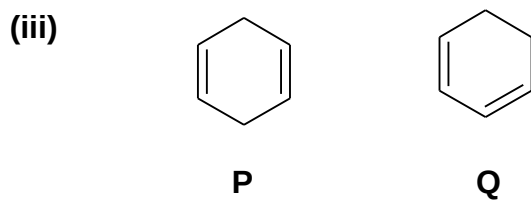
- (e) For each of the following pairs of compounds, suggest **one** simple chemical test to distinguish them from each other. State the reagents and conditions needed and give the expected observations for each compound.



[2]



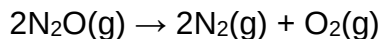
[2]



[2]

[Total:22]

- 3 Nitrous oxide, N_2O , is commonly known as laughing gas due to the euphoric effects of inhaling it. At a temperature of 1000 K, in the presence of chlorine catalyst, N_2O decomposes to its elements according to the following equation:



The experimental rate equation for this reaction is:

$$\text{Rate} = k P_{\text{N}_2\text{O}}^a P_{\text{Cl}_2}^b,$$

where $P_{\text{N}_2\text{O}}$ and P_{Cl_2} are the partial pressures of N_2O and Cl_2 respectively, and a and b are non-zero constants.

The rate of decomposition of pure N_2O may be followed at constant temperature, by measuring the total pressure of the system with time.

The data below refers to a reaction in which pure N_2O decomposes in the presence of chlorine catalyst, Cl_2 , (at partial pressure of 47 kPa).

Experiment 1:

time / s	0	40	80	140	240
P_{total} / kPa	87.0	91.3	93.0	98.0	102.2
$P_{\text{N}_2\text{O}}$	40	31.4	25	18	9.6

- (a) (i) Show that the partial pressure of N_2O at any instant is given by
 $P_{\text{N}_2\text{O}} = 214 - 2P_{\text{total}}$ [2]

- (ii) Use the given data provided in the table above to plot a graph of $P_{\text{N}_2\text{O}}$ against time on suitable axes.

Showing all your working and drawing clearly any construction lines on your graph, use your graph to determine the value of a , the order of reaction with respect to N_2O . [4]

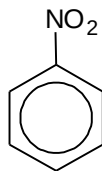
- (iii) Two additional experiments, experiments 2 and 3 were carried out and the following results were obtained.

	$P_{\text{N}_2\text{O}}$ / kPa	P_{Cl_2} / kPa	Initial rate/ kPa s ⁻¹
experiment 2	70	35	0.306
experiment 3	70	70	0.613

Determine b , the order of reaction with respect to chlorine. Explain your reasoning. [1]

- (iv) Hence write the rate equation for the reaction. [1]

- (b) Nitrobenzene is an organic compound used to mask unpleasant odors in shoe and leather dressings.



- (i) Suggest the type of hybridisation of N in nitrobenzene. [1]
- (ii) Nitrobenzene can be formed from reacting benzene with concentrated nitric acid and sulfuric acid. Name and outline the mechanism of the nitration of benzene. Indicate the rate-determining step and draw the formula of the organic intermediate. [3]
- (iii) Given that the nitration of benzene is exothermic, sketch the reaction pathway diagram of the reaction, labelling clearly the reactant, intermediate and product on your diagram. [2]
- (c) To investigate the mechanism of nitration of benzene, every hydrogen atom on benzene is replaced by its isotope, deuterium, to form C₆D₆.

Carbon–deuterium (C-D) bond has higher bond energy than a carbon–hydrogen (C-H) bond.

If the rate-determining step involves the breaking of a C–H bond, replacing the C–H bond with a C–D bond will change the rate constant, k , of the reaction. This is known as the deuterium isotope effect.

$$\text{Deuterium isotope effect} = \frac{k_H}{k_D}$$

k_H = rate constant for nitration of C₆H₆

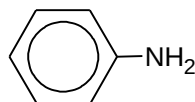
k_D = rate constant for nitration of C₆D₆

If $\frac{k_H}{k_D} = 1$, it is said that the deuterium isotope effect is absent.

If $\frac{k_H}{k_D} \gg 1$, it is said that the deuterium isotope effect is present.

Predict with reasoning, if the deuterium isotope effect would be present in the nitration of benzene. [1]

- (d) (i) Nitrobenzene can be converted to phenylamine, which is used in the synthesis of polyurethane, a polymer used in making tyres.



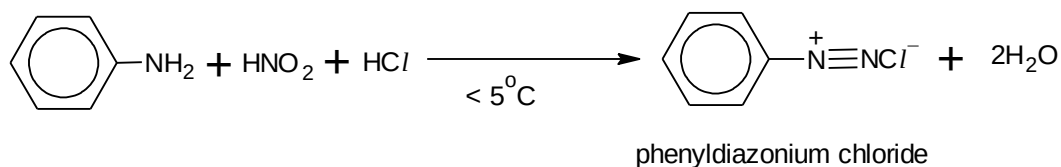
phenylamine

Suggest the reagents and conditions to convert nitrobenzene to phenylamine. [1]

- (ii) The following equations illustrate the formation of phenol from phenylamine, in two steps:

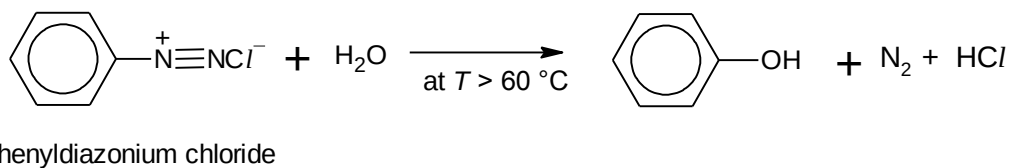
In step 1, phenylamine reacts with cold nitrous acid, HNO_2 , and hydrochloric acid, HCl , to form phenyldiazonium chloride.

step 1

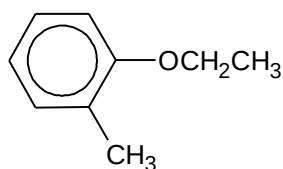


In step 2, phenyldiazonium chloride can react with water upon heating to give phenol.

step 2



Propose a two-step synthetic pathway for the conversion of 2-methylphenylamine to compound **S** below.

Compound **S**

[2]

[Total:18]

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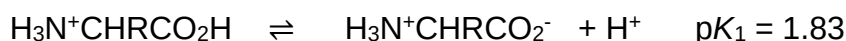
Section B

Answer **one** question from this section.

- 4 (a) A blood test can detect abnormally high levels of an amino acid which is one of the signs of the disease phenylketonuria, a rare condition in which a person is unable to properly break down amino acids such as phenylalanine.

In acidic solution, phenylalanine is completely protonated and exists as $\text{H}_3\text{N}^+\text{CHRCO}_2\text{H}$, where R is $-\text{CH}_2\text{C}_6\text{H}_5$.

Protonated phenylalanine, $\text{H}_3\text{N}^+\text{CHRCO}_2\text{H}$, acts as a dibasic acid that ionises in stages.



- (i) Calculate the pH of a 0.80 mol dm^{-3} solution of protonated phenylalanine (ignore the effect of pK_2 on the pH). [1]
- (ii) An amphiprotic species is one that can act as a Bronsted-Lowry acid or base. The pH of a solution containing an amphiprotic species is given by the following expression.

$$\text{pH} = \frac{1}{2}(pK_1 + pK_2)$$

In the titration of protonated phenylalanine with NaOH, an amphiprotic species is formed.

Identify the amphiprotic species formed and calculate the pH of the solution. [2]

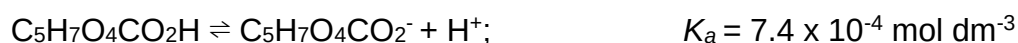
- (iii) Sketch the titration curve when 25.0 cm^3 of the protonated form of phenylalanine is being titrated with 70 cm^3 of NaOH(aq) of the same concentration. On your sketch, clearly mark the two pK_a values and the points you have calculated in (i) and (ii). [3]
- (iv) Suggest a suitable indicator from the following table to be used to detect the first equivalence point and state the colour change of the solution at this equivalence point.

indicator	pH range	acid solution	basic solution
Thymol Blue	1 – 3	red	yellow
Methyl Red	5 – 6	red	yellow
Phenol red	7 – 8	yellow	red
Phenolphthalein	8 - 10	colourless	red

[2]

- (b) “Acidity regulators” are food additives that have a buffering action on the pH of food. A mixture of citric acid, $\text{C}_5\text{H}_7\text{O}_4\text{CO}_2\text{H}$, and its sodium salt is often used for this purpose.

You may assume that citric acid behaves as a monobasic weak acid.



- (i) A citric acid / sodium citrate buffer mixture is prepared by mixing 40.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ citric acid and 60.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ sodium citrate.

Calculate the pH of the buffer solution. [1]

- (ii) Write equations to show how this mixture of citric acid and sodium citrate regulates the acidity on addition of H^+ ions and OH^- ions. [2]

- (c) (i) Describe the conditions needed to hydrolyse a protein non-enzymatically. [1]

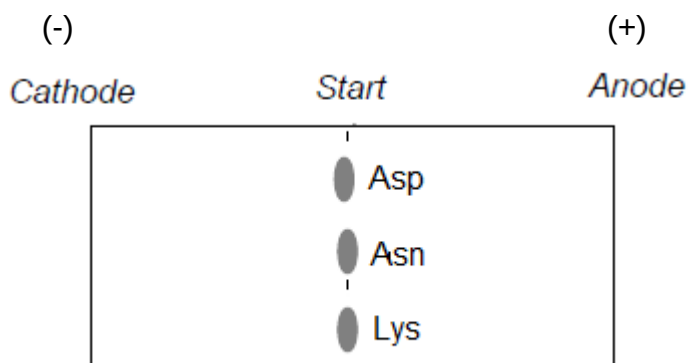
- (ii) Histidine decarboxylase undergoes enzymatic hydrolysis. Upon hydrolysis, the polypeptide fragments of the enzyme’s active site are isolated as shown below.

- Ala-Cys-Phe
- Gly-Gly
- Lys-Asp-Asp-Gly-Gly
- Phe-Arg-Lys

Deduce the sequence of the 9 amino acid residues of the enzyme’s active site. [2]

- (iii) A different segment of tripeptide, Asp-Asn-Lys, was subjected briefly to acidic hydrolysis which produced individual amino acids as well as various peptides. The resulting mixture buffered at pH 13 was separated in an electric field using electrophoresis.

Amino acid	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{COOH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{NH}_2 \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ (\text{CH}_2)_5 \\ \\ \text{NH}_2 \end{array}$
Abbreviation	Asp	Asn	Lys



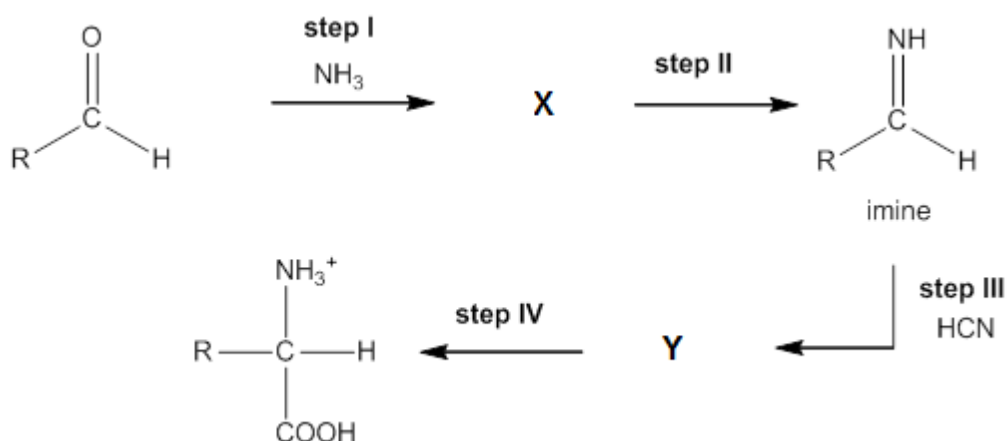
Using the above template, sketch and label the position of the three amino acids after the separation. State one factor that affects the rate at which the charged particles move during electrophoresis.

[3]

(d) The Strecker synthesis assembles an α -amino acid from the amine precursor and an aldehyde.

- Step I is a nucleophilic addition reaction.
- Step II involves the elimination of water from **X** to form an imine.
- Step III is similar to the formation of cyanohydrin when the imine undergoes nucleophilic addition with HCN and forms an α -amino nitrile, **Y**.
- Step IV is the hydrolysis of **Y** to form the carboxylic acid functional group.

In the following reaction flow scheme, draw the intermediate compounds **X** and **Y** and suggest the reagent and condition for step IV in the scheme.



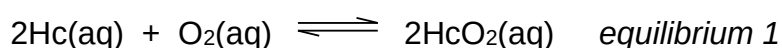
[3]

[Total:20]

- 5 Copper is an important transition element that can readily form complex ions with *ligands*. Copper forms a large variety of compounds, usually with variable oxidation states. Copper compounds also act as useful *homogeneous catalysts*, commonly used in many industrial operations.

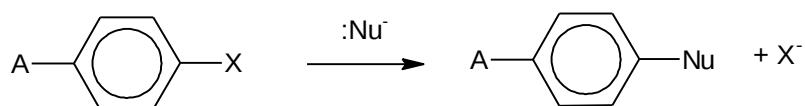
- (a) Define the terms *ligand* and *homogeneous catalyst*. [2]
- (b) In recent years, horseshoe crabs have generated great interest amongst scientists due to its blood with remarkable antibacterial properties. An unusual aspect of the blood is that it is bright blue, a consequence of using copper-based complexes, haemocyanin, to transport oxygen.

Two Haemocyanin (Hc) reversibly bind a single oxygen molecule, O₂, to form oxyhaemocyanin, HcO₂, according to the following equilibrium:



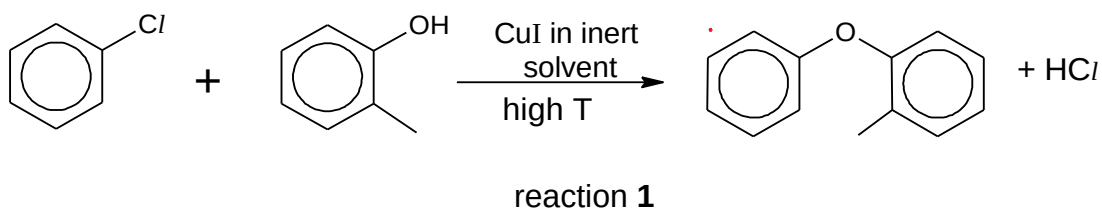
Oxygenation causes a color change between Cu(I) deoxygenated form and the Cu(II) oxygenated form.

- (i) By considering *equilibrium 1* and the level of oxygen in both the lungs and body cells, comment on how the crab cells receive oxygen from the air in the lungs. [2]
- (ii) Indicate the color change from Cu(II) oxygenated form to Cu(I) deoxygenated form. Hence explain why both copper complex ions exhibit different colours. [5]
- (c) Most aromatic compounds undergo electrophilic substitution. However it is possible for aryl halides to undergo a limited number of nucleophilic substitution reactions with strong nucleophiles. An example of a reaction is as follows:

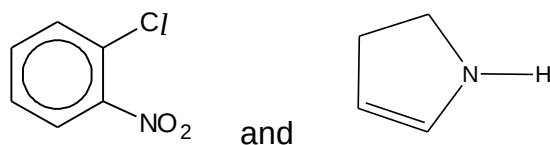


An example of an aryl nucleophilic substitution is the *Ullmann coupling* reaction where hydrocarbons fragments are coupled, using copper compounds as a catalyst.

An example is indicated in reaction 1 below.

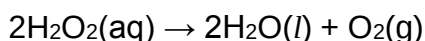


- (i) The conditions for the *Ullmann coupling* reactions are usually harsh conditions, and yet give low yield. In a similar reaction under the same temperature and pressure, the two compounds below were reacted together:



Deduce and draw the structure of the product for the above reaction. [1]

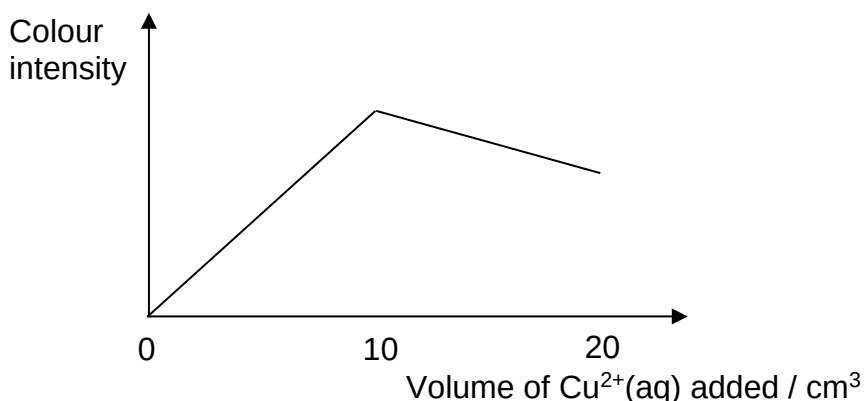
- (ii) Predict, with reasoning, if the above reaction in (c)(i) will result in a higher or lower yield compared to reaction 1. [2]
- (iii) Like copper(I) iodide, aqueous Fe^{2+} is a good homogeneous catalyst used in the decomposition of aqueous hydrogen peroxide as indicated below.



With reference to relevant data in the *Data Booklet*, suggest a mechanism for the catalysis of the above reaction by aqueous Fe^{2+} . [3]

- (d) Copper(II) species can have various coordination numbers.

The formula of a methylamine-copper complex ion, $[\text{Cu}(\text{CH}_3\text{NH}_2)_x(\text{H}_2\text{O})_2]^{2+}$, can be determined using a colorimeter. In an experiment, a solution of $0.050 \text{ mol dm}^{-3}$ of $\text{Cu}^{2+}(\text{aq})$ was added 1 cm^3 at a time to 20 cm^3 of $0.100 \text{ mol dm}^{-3}$ of aqueous CH_3NH_2 . The colour intensity of the resultant solution after each addition was measured using a colorimeter. The following graph was obtained.



- (i) Using the graph, determine the value of x . [2]
- (ii) Draw the structure of the methylamine-copper complex ion. [1]

- (e) Mercury is poisonous as it can cause many health problems ranging from neurological, respiratory to cardiovascular problems. One possible treatment for mercury poisoning involves administering a solution of EDTA^{4-} , a common hexadentate ligand which forms complexes with many metal ions.

Copper and calcium ions are essential minerals required by the body. Using relevant data from below, comment on the use of EDTA^{4-} as a treatment for mercury poisoning and comment on its effect on the concentrations of copper and calcium ions in the body.

Equilibrium				$K_c / \text{mol}^{-1} \text{dm}^3$
Ca^{2+}	+	$\text{EDTA}^{4-} \rightleftharpoons$	$[\text{Ca}(\text{EDTA})]^{2-}$	6×10^{10}
Cu^{2+}	+	$\text{EDTA}^{4-} \rightleftharpoons$	$[\text{Cu}(\text{EDTA})]^{2-}$	5×10^{18}
Hg^{2+}	+	$\text{EDTA}^{4-} \rightleftharpoons$	$[\text{Hg}(\text{EDTA})]^{2-}$	6.3×10^{21}

[2]

[Total:20]

End of Paper